AetherGrid AI – UML Deployment Diagram

Nodes, execution environments, artifacts and communication paths of the complete system (implemented and planned)

«device»
Operator Workstation

«execution environment»
Web browser (HTML5 Canvas)

«artifact»
Mission Control dashboard
index.html, app.js, styles.css: 60 fps canvas, telemetry, node inspector, strategy selector

«artifact»
Scenario Laboratory panel
inject closures, hazards, weather, demand bursts and new emergency requests

«HTTP» REST / JSON

«WebSocket» state diff, 10 Hz

«device»
GitHub (cloud)

«artifact»
Source repository (Git)
core/, ai/, web/, tests/ with history and pull requests

«execution environment»
GitHub Actions runner

«artifact» (planned)
CI workflow
unit, property and invariant tests on every push

«artifact» (planned)
Benchmark regression gate
seeded runs; merge blocked beyond a ±5 % regression

«HTTPS / SSH» git push, clone

«device»
Simulation Host

«execution environment»
Python 3.11 + NumPy (isolated venv, standard library, no web framework)

«artifact»
web/server.py
hand-rolled HTTP server: REST API (route, closure, hazard, weather, surge, reset) and static dashboard

«artifact»
core/ simulation engine
grid, agents, incidents, event loop, seeded random generator, SimulationStateManager

«artifact»
core/ dynamic cost field
CostPricingEngine, HazardManager, CongestionManager (EWMA), WeatherManager (Markov)

«artifact»
ai/pathfinding.py
dynamic A*, Dijkstra ETA matrices, bidirectional and weighted A*; HPA* (planned)

«artifact» (planned)
ai/dispatchers.py
greedy and nearest-vehicle baselines, Hungarian batch matcher, genetic-algorithm VRPTW solver

«artifact» (planned)
ai/rl_agent.py
tabular Q-learning for idle-vehicle repositioning; optional DQN extension

«artifact» (planned)
web/ws_server.py
WebSocket streamer: 10 Hz state diffs to the dashboard

«artifact»
HazMat directive module
PPE protocol levels, hospital decontamination alerts, route exposure check

«artifact»
cli.py and tests/
serve, run, train, benchmark; unit, property and invariant tests; VerificationSuite

«file I/O» maps, logs, models

«device»
Local file system

«artifact»
data/maps/*.osm
preset and imported OpenStreetMap road networks, plus synthetic 24 x 24 grid

«artifact»
results/
benchmark JSON reports and SHA-256-verified deterministic metric logs

«artifact» (planned)
models/
Q-table and DQN weights saved by the train subcommand

Node (device)

Execution environment

Implemented artifact

Planned artifact

HazMat safety artifact

Communication path

Planned path